

# Sequential Multi-Target Delivery under Partial Observability

CA6126 Final Project Report

Group TBD

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## 1 Introduction

We study a reinforcement learning problem in which a delivery robot must visit  $K$  landmarks on an  $N \times N$  grid in a *hidden* order that is randomised at every episode. The agent receives only binary feedback (correct / wrong) after each visit attempt, and must therefore *infer* the ordering from its interaction history. This makes the task a Partially Observable Markov Decision Process (POMDP), where the optimal policy is inherently non-Markov in the observation and requires memory.

The problem is inspired by real-world scenarios such as multi-stop delivery routing where the priority of stops is unknown to the driver and can only be discovered on arrival. It is not a Travelling Salesman Problem: the order is fixed but unknown, and the agent must trade off *information gain* against *travel cost*.

We implement the environment as a Gymnasium-compatible Python module and train agents using the Stable Baselines 3 ecosystem. Our key contribution is a **systematic investigation of why standard RL algorithms fail on this POMDP** and the design choices needed to enable successful learning: reward shaping, penalty tuning, and problem scaling.

## 2 Problem Formulation

### 2.1 POMDP Specification

We formalise the problem as a POMDP tuple  $(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, R, O, \gamma)$ :

**Hidden state**  $s_t$ .

$$s_t = (p_t, \{\ell_i\}_{i=1}^K, \pi, k_t, t)$$

where  $p_t \in \{0, \dots, N-1\}^2$  is the robot position,  $\ell_i$  are landmark positions,  $\pi \in S_K$  is the hidden visiting permutation,  $k_t$  is the number of correct visits so far, and  $t$  is the step counter.

**Observation**  $o_t$ . The agent observes:

- A  $5 \times 5$  ego-centric view (one-hot encoding over  $\{\text{empty}, \text{landmark}_1, \dots, \text{landmark}_K, \text{wall}\}$ ).
- A history vector  $h_t \in \{-1, 0, +1\}^K$ :  $-1$  = tried & rejected,  $0$  = unknown,  $+1$  = confirmed correct.
- Progress  $k_t/K$  and normalised remaining steps  $(T - t)/T$ .

Crucially, the permutation  $\pi$  is **never** revealed.

**Action space.**  $\mathcal{A} = \{\text{Up}, \text{Down}, \text{Left}, \text{Right}, \text{Visit}\}$ ,  $|\mathcal{A}| = 5$ .

| Event                                | Reward               |
|--------------------------------------|----------------------|
| Per-step cost (any action)           | −0.01                |
| Wall bump                            | −0.05                |
| Visit on empty cell                  | −0.05                |
| Correct landmark visit               | +1.00                |
| Wrong landmark visit                 | − $p$ (configurable) |
| Completing full sequence ( $k = K$ ) | +5.00                |
| Timeout ( $t = T$ )                  | −1.00                |

Table 1: Reward function design. The wrong-visit penalty  $p$  is a key hyperparameter investigated in this work (default  $p = 0.5$ ).

### Reward structure.

## 2.2 Why This Is Novel

- Standard gridworlds expose the goal; here the goal-ordering is hidden.
- Unlike TSP, the order is fixed but unknown—pure shortest-path heuristics fail.
- The optimal policy requires *sequential hypothesis testing*: the agent must actively explore to narrow down the  $K!$  possible orderings.
- The history vector  $h_t$  is a sufficient statistic for belief tracking, but learning to *use* it is non-trivial.

**Reward shaping.** We implement a potential-based shaping term (Ng et al., 1999):

$$F(s, s') = \gamma \Phi(s') - \Phi(s), \quad \Phi(s) = -\frac{d_{\min}(p, \text{plausible landmarks})}{2(N-1)}$$

where  $d_{\min}$  is Manhattan distance to the nearest landmark with  $h_i = 0$  (untried). This gives a continuous positive gradient toward landmarks without changing the optimal policy in theory.

## 3 Environment Implementation

The environment is implemented as a Gymnasium `Env` subclass (`env.py`) using the Stable Baselines 3 framework for all algorithm implementations. Key design decisions:

1. **Dict observation space:** Separate ego-view, history, progress, and time budget into distinct keys for SB3’s `MultiInputPolicy`.
2. **Deterministic dynamics:** Stochasticity comes only from the hidden permutation and initial placement.
3. **Configurable difficulty:** Grid size ( $N$ ), number of landmarks ( $K$ ), view size, wrong-visit penalty ( $p$ ), and reward shaping are all adjustable.
4. **Headless rendering:** Pygame-based renderer (`render.py`) produces RGB frames for video recording.

## 4 Algorithms

All algorithms are from the Stable Baselines 3 (SB3) ecosystem:

- **PPO:** Feed-forward `MultiInputPolicy` with  $[256, 256]$  MLPs. 16 parallel envs.
- **Recurrent PPO (LSTM):** `MultiInputLstmPolicy` from `sb3-contrib` with 128-unit LSTM +  $[128, 128]$  MLPs. The LSTM maintains internal state across timesteps, enabling belief tracking.

- **DQN:** [256, 256] MLP with experience replay (200k buffer),  $\varepsilon$ -greedy exploration.
- **QR-DQN:** Distributional DQN with 51 quantiles.

## 5 Experimental Journey: From Failure to Success

This section documents our iterative debugging process, which is central to understanding how to solve POMDPs with deep RL.

### 5.1 Phase 1: Large Grid Baseline ( $12 \times 12$ , $K=4$ , $p=0.5$ )

**Setup.** Our initial configuration followed standard practice:  $12 \times 12$  grid, 4 landmarks ( $4! = 24$  orderings),  $5 \times 5$  ego-view, wrong-visit penalty  $p = 0.5$ , no reward shaping. We trained PPO, DQN, and RecurrentPPO for 5M steps each.

| Algorithm       | Reward           | Success Rate | Wrong Visits | Ep. Length |
|-----------------|------------------|--------------|--------------|------------|
| Random baseline | $-5.95 \pm 0.94$ | 0.0%         | 0.99         | 200.0      |
| PPO (3 seeds)   | $-3.00 \pm 0.00$ | 0.0%         | 0.00         | 200.0      |
| DQN (seed 0)    | $-3.97 \pm 7.67$ | 0.0%         | 1.73         | 200.0      |
| RecurrentPPO    | $-3.00 \pm 0.00$ | 0.0%         | 0.00         | 200.0      |

Table 2: Phase 1: All algorithms fail on the large grid. 200 eval episodes.

#### Results.

**Analysis: The “Never Visit” Local Optimum.** Every algorithm converged to a policy that *never executes the Visit action*. The reward decomposition reveals why:

$$\underbrace{200 \times (-0.01)}_{\text{step cost}} + \underbrace{(-1.0)}_{\text{timeout}} = -3.0$$

Meanwhile, each exploratory visit attempt has **negative expected value**:

$$\mathbb{E}[\text{single visit}] = \frac{1}{4} \times (+1.0) + \frac{3}{4} \times (-0.5) = -0.125$$

The agent rationally concludes that *doing nothing* ( $-3.0$ ) beats *trying* (expected  $< -3.0$ ). This is a fundamental exploration problem: the large positive reward from completing the sequence ( $+8.0$ ) is too unlikely to discover through random exploration.

#### Additional factors.

- **Low visibility:**  $5 \times 5$  view covers only  $25/144 = 17\%$  of the  $12 \times 12$  grid. Landmarks are usually invisible.
- **High combinatorial complexity:**  $4! = 24$  orderings require extensive exploration to disambiguate.
- DQN performs worst ( $-3.97$ , 1.73 wrong visits) because  $\varepsilon$ -greedy forces random Visit actions, incurring penalties.

### 5.2 Phase 2: Adding Reward Shaping ( $12 \times 12$ , $K=4$ , $p=0.5$ , shaped)

**Hypothesis.** The agent cannot discover landmarks because it has no gradient signal pointing toward them. Potential-based reward shaping should guide navigation without changing the optimal policy.

**Results.** RecurrentPPO with shaping reached train reward  $-2.69$  at 2M steps (vs.  $-3.03$  without shaping). The improvement indicates the agent learned to **approach landmarks**, but eval reward remained  $-3.00$  with 0% success—the best model saved by EvalCallback still corresponded to the “never visit” strategy because eval uses unmodified rewards.

**Takeaway.** Reward shaping helps navigation but is insufficient alone: the agent approaches landmarks but still does not dare to visit them.

### 5.3 Phase 3: Reducing Wrong-Visit Penalty ( $12 \times 12$ , $K=4$ , $p=0.1$ , shaped)

**Hypothesis.** With  $p = 0.1$ , a single visit attempt has positive expected value:

$$\mathbb{E}[\text{single visit}] = \frac{1}{4} \times (+1.0) + \frac{3}{4} \times (-0.1) = +0.175 > 0$$

This should incentivise exploration.

**Results.** RecurrentPPO with shaping + low penalty reached  $-2.62$  at 1.3M steps, slightly better than Phase 2 at the same stage. However, eval success rate remained 0%—the  $12 \times 12$  grid is still too large for the agent to reliably find and visit landmarks.

**Takeaway.** Lowering penalty makes exploration rational in expectation, but spatial exploration on a large grid with limited vision remains the bottleneck.

### 5.4 Phase 4: Scaling Down ( $6 \times 6$ , $K=3$ , $p=0.1$ , shaped) — Breakthrough

**Hypothesis.** The core POMDP challenge (inferring hidden order) is orthogonal to the spatial navigation challenge. By reducing the grid to  $6 \times 6$  (view coverage:  $25/36 = 69\%$ ), landmarks to 3 (only  $3! = 6$  orderings), and episode length to 100 steps, we isolate the POMDP reasoning from spatial difficulty.

| Configuration                   | Steps | Reward           | Success      | Wrong Visits | Ep. Len |
|---------------------------------|-------|------------------|--------------|--------------|---------|
| Random ( $6 \times 6$ , $K=3$ ) | —     | $\sim -3.0$      | 0%           | $\sim 1.0$   | 100     |
| RecurrentPPO ( $6 \times 6$ )   | 1.5M  | $+6.09 \pm 3.04$ | <b>87.5%</b> | low          | $< 100$ |

Table 3: Phase 4: Small grid breakthrough. RecurrentPPO achieves 87.5% success rate. Eval over 200 episodes.

**Results.**

**Training dynamics.** The train reward trajectory tells a clear story of learning progression:

- **0–50k steps:** Random exploration, reward  $\sim -3.15$ .
- **50k–200k:** Agent learns navigation toward landmarks ( $-3.15 \rightarrow -2.59$ ), driven by reward shaping.
- **200k–800k:** Agent begins attempting visits, reward improves through correct deliveries ( $-2.59 \rightarrow -1.87$ ).
- **800k–1.5M:** LSTM learns to track history and plan multi-step visit sequences ( $-1.87 \rightarrow -1.35$  train, eval reward best  $-1.68$ , eval success rate 87.5%).

**Best episode reward:**  $+7.91$ . This is close to the theoretical maximum of  $3 \times 1.0 + 5.0 - \text{steps} \times 0.01 = +7.0 \sim +7.9$ , confirming the agent learned the near-optimal strategy.

## 6 Discussion

### 6.1 Why Each Design Change Mattered

| Change                                            | Best Reward   | Success      | What it solved                |
|---------------------------------------------------|---------------|--------------|-------------------------------|
| Baseline ( $12 \times 12$ , $p=0.5$ , no shaping) | -3.00         | 0%           | —                             |
| + Reward shaping                                  | -2.69 (train) | 0%           | Navigation toward landmarks   |
| + Low penalty ( $p=0.1$ )                         | -2.62 (train) | 0%           | Made exploration rational     |
| + Small grid ( $6 \times 6$ , $K=3$ )             | +6.09         | <b>87.5%</b> | Spatial coverage & complexity |

Table 4: Ablation summary: each design change addresses a specific failure mode.

### 6.2 The Three Barriers to POMDP Learning

Our experiments reveal three independent barriers that must all be overcome:

1. **Exploration incentive:** The expected reward of a single visit attempt must be positive ( $p < 1/K \times r_{\text{correct}} / (1 - 1/K)$ ). With  $K = 4$  and  $r_{\text{correct}} = 1.0$ , this requires  $p < 0.33$ . Our original  $p = 0.5$  violated this, making “never visit” the rational policy.
2. **Spatial coverage:** The agent’s ego-view must cover enough of the grid that landmarks are frequently visible. On a  $12 \times 12$  grid with  $5 \times 5$  view (17% coverage), the agent spends most of its time navigating blind. On  $6 \times 6$  (69% coverage), landmarks are almost always visible.
3. **Memory capacity:** The policy must maintain beliefs over  $K!$  orderings. RecurrentPPO (LSTM) succeeds where PPO (MLP) fails, even with the hand-crafted history vector—confirming that **learned memory** is essential for this POMDP.

### 6.3 Feed-Forward vs. Recurrent: A POMDP Litmus Test

The stark difference between PPO (0% success, all configurations) and RecurrentPPO (87.5% on the feasible configuration) demonstrates that this environment is a **genuine POMDP** that cannot be solved by Markov approximations. The history vector  $h_t$  is a sufficient statistic in theory, but the MLP-based PPO cannot learn to use it effectively—the LSTM discovers its own internal representation that better supports sequential decision-making.

### 6.4 Scaling Considerations

Our results suggest that the  $12 \times 12$ ,  $K = 4$  configuration is solvable in principle but requires:

- Larger view size (e.g.,  $7 \times 7$  or  $9 \times 9$ ) or global position information in the observation.
- Longer training (possibly  $>50\text{M}$  steps) with curriculum learning.
- Larger LSTM capacity and possibly attention-based architectures.

### 6.5 Limitations

- We only achieved success on the  $6 \times 6$  configuration; the  $12 \times 12$  version remains unsolved.
- The EvalCallback saves models based on unmodified reward, so shaped training + unmodified eval creates a mismatch where “never visit” appears optimal to the callback.
- Only 3 seeds were run; larger seed counts would give tighter confidence intervals.

## 7 Conclusion

We presented Sequential Multi-Target Delivery, a novel POMDP environment where a robot must infer a hidden visiting order through trial-and-error. Through systematic experimentation, we identified three barriers to learning—exploration incentive, spatial coverage, and memory capacity—and showed that all three must be addressed simultaneously.

Our main findings:

1. **Feed-forward methods fail:** PPO and DQN converge to a “never visit” avoidance strategy ( $-3.0$  reward, 0% success) across all configurations tested.
2. **Reward shaping + penalty tuning are necessary but not sufficient:** They improve training reward but do not achieve task completion on a large grid.
3. **RecurrentPPO succeeds** when the problem is scaled to a feasible difficulty:  $6 \times 6$  grid,  $K = 3$  landmarks, with reward shaping and low wrong-visit penalty. It achieves **87.5% success rate** and near-optimal reward ( $+6.09$ , best  $+7.91$ ).
4. **The iterative debugging process**—from total failure to 87.5% success—demonstrates the importance of systematic ablation in RL research.

## Appendix: Hyperparameters

| Parameter                   | PPO                | RecurrentPPO               | DQN                                    | QR-DQN                                 |
|-----------------------------|--------------------|----------------------------|----------------------------------------|----------------------------------------|
| Policy network              | [256, 256] MLP     | [128, 128] MLP + LSTM(128) | [256, 256] MLP                         | [256, 256] MLP                         |
| Learning rate               | $3 \times 10^{-4}$ | $3 \times 10^{-4}$         | $2.5 \times 10^{-4}$                   | $2.5 \times 10^{-4}$                   |
| Batch size                  | 512                | 256                        | 128                                    | 128                                    |
| $n_{\text{steps}}$ / buffer | 512                | 256                        | 200k                                   | 200k                                   |
| Discount $\gamma$           | 0.99               | 0.99                       | 0.99                                   | 0.99                                   |
| Parallel envs               | 16                 | 16                         | 1                                      | 1                                      |
| $n_{\text{epochs}}$         | 5                  | 5                          | —                                      | —                                      |
| Exploration                 | ent. coef. 0.01    | ent. coef. 0.01            | $\varepsilon$ : 1.0 $\rightarrow$ 0.05 | $\varepsilon$ : 1.0 $\rightarrow$ 0.05 |
| Total steps                 | 5M                 | 2–5M                       | 5M                                     | 5M                                     |

Table 5: Hyperparameter summary.

| Env Parameter           | Large (Phase 1–3) | Small (Phase 4) |
|-------------------------|-------------------|-----------------|
| Grid size $N$           | 12                | 6               |
| Landmarks $K$           | 4                 | 3               |
| Orderings $K!$          | 24                | 6               |
| View size               | $5 \times 5$      | $5 \times 5$    |
| View coverage           | 17%               | 69%             |
| Max steps $T$           | 200               | 100             |
| Wrong-visit penalty $p$ | 0.5               | 0.1             |
| Reward shaping          | varies            | Yes             |

Table 6: Environment configurations across experimental phases.